Tejas Sonwane

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EDUCATION

MSc Statistics & Data Science **2025 – Present**
Nilkamal School of Mathematics, Applied Statistics and Analytics, SVKM's NMIMS, Mumbai CGPA – 7.61/10

B.Sc Statistics **2022 – 2025**
KTHM College, Nashik CGPA – 8.30/10

Senior Secondary Examination (HSC) – 12th **2019 – 2021**
KTHM College, Nashik Percentage – 81.67%

Secondary Examination - 10th (SSC) **April, 2019**
New Maratha High School, Nashik Percentage – 78.60%

INTERNSHIP

Adbureau Analytics Private Limited **May 2026 – Present**
Business & Technology Analyst

- Built an ML model on real-world clinical data of patients on diabetes, applying class imbalance handling, hyperparameter tuning, feature engineering, and multiple classifiers and ensemble strategies, evaluated using recall, precision, F1, and ROC-AUC, achieving 82% recall on the final stacking ensemble.
- Applied SHAP explainability to surface key clinical risk drivers and delivered data-driven business recommendations to improve hospital discharge outcomes.
- Built a multi-agent AI system (Discovery, Profiler, and Scoring & Outreach) using LangChain and the Google Gemini API to autonomously identify, research, and score 400+ real-world business leads for targeted outreach.
- Designed and maintained a PostgreSQL (Supabase) data pipeline with automated retry handling, deduplication, and confidence-scoring, enabling reliable end-to-end processing across 30+ product categories. Built a FastAPI-backed dashboard and Docker packaging to improve usability and portability.

PROJECTS

Predictive Credit Risk Modelling using Statistical Techniques **Jan 2025–Mar 2025**
Methods: Logistic Regression(GLM), Random Forest, XGBoost(GBM), Hypothesis Testing (Chi-Square, Mann-Whitney U),EDA

- Conducted end-to-end credit risk analysis on a 255K+ record loan dataset, performing exploratory data analysis and hypothesis testing (Chi-Square, Mann-Whitney U with effect sizes) to identify key predictors of default age, interest rate, employment tenure, and income, and validated findings against a real-world benchmark dataset (UCI German Credit) to assess data reliability.
- Built and compared five classification models (Logistic Regression, Random Forest, XGBoost) with class-imbalance handling (class-weighting, SMOTE) and threshold tuning, used SHAP for model interpretability and feature-level explainability.

Survival Analysis of Diabetic Patient Data Using X-Lindley Distribution **July 2025-Nov 2025**
Methods:MLE, Survival Analysis, Kaplan-Meier, Kolmogorov–Smirnov Test | R

- Built a statistical survival model on 330 diabetic patient records, comparing 4 probability distributions (X-Lindley, Lindley, Exponential, Zeghdoudi) using MLE and MoM to identify the best-fitting model for time-to-event data
- Ran Chi-Square and Kolmogorov-Smirnov goodness of fit tests to check which distribution best fits the data, where X-Lindley was the only one that passed the K-S test with a statistic of 0.0729 against a critical value of 0.0903 at 1% significance level, confirming it as the best fit
- Built Kaplan-Meier survival curves in R stratified by diabetes status, where survival probability dropped from 61% at age 40 to 43% by age 50, identifying age 40 to 42 as the point of sharpest survival decline for diabetic patients

Hybrid Feature Selection & ML-Based Cancer Classification **Dec 2025-Mar 2026**
Methods:Hybrid Feature Selection, Genetic Algorithm | Python (scikit-learn)

- Worked on gene expression data from four cancer types (Brain, Breast, Lung, CNS) on microarray datasets with up to 24,481 gene features and 279 patient samples.
- Built a two-step pipeline to cut down the number of genes: used statistical filters (IG, Gain Ratio, Chi-Squared) to remove weak genes first, further picking the best genes combination and reducing the gene count by 97% using Genetic Algorithm
- Trained and tested 7 machine learning models (SVM, Random Forest, Decision Tree, KNN, Naive Bayes) on the reduced gene set using cross-validation, and compared them on evalution metrics to identify the best-performing model per dataset

CERTIFICATIONS

- Certificate Course in Data Analysis – SPPU/UGC STRIDE (2022–23)
- Employability Enhancement Programme – Skill Up, Infosys Foundation (2025)
- SQL - MySQL for Data Analytics & Business Intelligence Udemy (2025-26)

POSITION OF RESPONSIBILITY

Discussion Club Head **Sept 2025 – Present**
Analytics Cell NSoMASA

- Led Discussion Club, organizing structured debates and discussion sessions to build critical thinking and public speaking skills across 120+ students while fostering peer-led leadership & coordinated Analytics Cell operations including hackathon team supervision, managing Sankhiyiki (annual statistics fest), quizzes, and workshops to drive end-to-end event execution.

TECHNICAL SKILLS

SQL | Python (Pandas, NumPy, Scikit-learn) | R | Excel | Tableau | Power BI | GitHub